The Kaskawulsh Glacier

A vast glacier in Canada that is notable for its sheer size and impressive medial moraines

Around 400 square miles (1,000 square km) in area, the Kaskawulsh Glacier snakes for some 45 miles (70 km) through the St. Elias Mountains of Canada's Yukon Territory. At its terminus, or snout, it produces meltwater that maintains the level of one of the Yukon's largest lakes, Kluane Lake.

Bold stripes

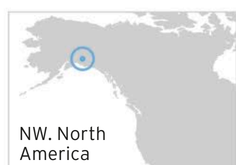
All glaciers constrained by valleys have dark bands of material running along their sides, called lateral moraines (see panel, p.150). These contain rock fragments that have been plucked off the walls of the valley and are being carried along by the glacier. When two glaciers merge to form a larger glacier, the lateral moraines of the smaller glaciers typically combine to form a bold dark stripe running down the merged glacier. This stripe is called a medial moraine. The lower half of the Kaskawulsh Glacier has several bold medial moraines, formed through a series of minimergers of its different arms, or tributaries, upstream. The glacier has three main arms, each with its own tributaries.

The Kaskawulsh's **broadest medial moraine** is over **1,300 ft (400 m) wide**

▷ **MERGER OF EQUALS**
In this view, different arms of the glacier can be seen to merge in the distance. The merger produces the widest medial moraine (dark band) on the glacier's downstream surface.

The Malaspina Glacier

The world's largest and finest example of a piedmont glacier—a huge lobe of ice spread out over a broad, flat plain



The enormous Malaspina Glacier is up to 40 miles (65 km) wide and extends up to 28 miles (45 km) forward from where its ice "root" emerges from between a gap in some mountains near the coast of Alaska. Studies of earthquake waves passing through it have shown that its ice is up to 2,000 ft (600 m) thick.

Patterned lobe

The glacier has complex patterns of light and dark bands on its surface—the dark areas are moraines, containing rock debris. The bands are arranged in zigzags and swirls, and are thought to have been caused by alternating, perhaps seasonal, patterns of fast and slow ice flow. Also on the surface are colonies of tiny creatures called ice worms.

△ **ICE SWIRL**
The glacier, with its vast swirling ice patterns, is named after an Italian explorer, Alessandro Malaspina, who visited it in 1791.

PIEDMONT GLACIERS

A piedmont glacier occurs where the flow of ice within a valley spills out through a narrow opening onto a broad plain, often near a coast. The ice spreads out to form a wide lobe.